

Exponential Relaxation of Fisher Path Speed on the Statistical Manifold of Learned Transition Matrices

Numerical Evidence from Quantum Processors

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Continuation of Grammar Fingerprinting (doi:10.5281/zenodo.19158088), Fisher Information Threshold Study (doi:10.5281/zenodo.19391582), LLM Grammar Fingerprinting (doi:10.5281/zenodo.19454201), and Curvature at the Fisher Threshold (Research Note, April 2026).

Abstract

We study the dynamics of learned transition matrices on the Fisher–Rao statistical manifold as a function of data length N , using the Grammar Fingerprinting pipeline applied to 28 Google Sycamore quantum processor readout configurations. By computing the Fisher path speed—the rate of change of the transition matrix $T(N)$ on the manifold, measured via the Fisher information metric—we show that the speed decays exponentially, $v(N) = v_\infty + A \exp(-N/\tau)$. The exponential model is preferred (lower AIC) in 25–26 of 28 readouts for each alphabet size $K \in \{5, 7, 9\}$ (26/28 at $K=5$, 25/28 at $K=7$, 25/28 at $K=9$). The relaxation timescale τ is systematically larger than the Fisher trace threshold N^* by $\tau/N^* \approx 3.2$ (median). Within each fixed K , the product $A \times \tau$ is stable across hardware topologies among exponential-fit readouts: at $K=7$ topology-wise medians agree within $\sim 2\%$ and topology-level CV ranges from 0.13 (Bulk) to 0.23 (Snake; Table 1); qualitatively similar patterns hold at $K=5$ and $K=9$ (CV ~ 0.07 – 0.22 on $A \times \tau$ within topology). Across $\{5, 7, 9\}$, medians of $A \times \tau$ follow $A \times \tau \approx 21.5 K^{0.57}$ with residuals below 1.4%. We compare the Fisher speed profile with the macroscopicity threshold of Coppo, Pranzini and Verrucchi (2022): both exhibit exponential decay with logarithmic threshold dependence on a tolerance parameter. The Grammar Fingerprinting threshold N^* and the Coppo threshold N_t are identified as two distinct points on the same exponential curve—structure onset ($\sim 27\%$ decay) versus classical completion ($\sim 95\%$ decay). The formal derivation of this correspondence remains an open problem.

1 Introduction

The Grammar Fingerprinting methodology (Csáplár, 2026a) established that quantum processor readout sequences carry statistically learnable temporal structure, detectable via the Fisher information trace of a SAX-discretized, LSTM-learned transition matrix. The Fisher Information Threshold Study (Csáplár, 2026b) showed that a data-length threshold N^* exists below which no reliable structure is detectable, and that this threshold is consistent across 28 Sycamore readout configurations, 3 LSTM seeds, and broadly aligned with the $\sim 8,000$ -point classification threshold from supervised experiments. The LLM extension (Csáplár, 2026c) demonstrated domain agnosticism: the same pipeline detects temporal grammar in large language model entropy series with signal-to-noise ratios of 2.2 – $263\times$. The Curvature Research Note (Csáplár, 2026d) introduced a second characteristic scale N_{curv} (the maximum of $|d^2\text{Tr}(F)/dN^2|$) and conjectured a geometric phase transition on the statistical manifold at the threshold.

All prior work analyzed the Fisher trace $\text{Tr}(F)$ —a scalar summary of the learned transition matrix $T(N)$. The present study moves beyond scalars to the geometry of the full transition matrix manifold. We compute the saved $T(N)$ matrices for all (readout, N) pairs, embed them in the Fisher–Rao manifold of row-stochastic matrices, and measure the Fisher path speed: the norm of the manifold velocity dT/dN , evaluated in the Fisher information metric. This provides a direct, geometry-aware measure of how rapidly the learned grammar changes as data accumulates.

We find that the Fisher path speed decays exponentially in N , with a characteristic timescale τ that is robust across topologies. The product $A \times \tau$ (amplitude times timescale) is an empirical summary within a given K , and scales approximately as a power law across K . We compare this relaxation profile with the macroscopicity threshold of Coppo, Pranzini and Verrucchi (2022), who derived the minimum system size N_t for the emergence of classical-like behavior in the Page–Wootters mechanism via generalized coherent state discrimination.

2 Methods

2.1 Transition matrix extraction and storage

The Grammar Fingerprinting pipeline (SAX alphabet K , LSTM `hidden_dim` = 32, `seq_len` = 16, `epochs` = 50) is applied to each of the 28 Sycamore readout configurations on a data-length grid of 15 points ($N = 500, 1000, 2000, \dots, 40000$). For each (readout, N) pair, the $K \times K$ row-stochastic transition matrix $T(N)$ is saved, yielding 420 matrices per K value. Three alphabet sizes are tested: $K \in \{5, 7, 9\}$.

Here $d = K(K - 1)$ counts the total number of free coordinates in the row-stochastic model: each of the K rows lies on a $(K - 1)$ -simplex, and the Fisher–Rao metric is taken with respect to the product of these K simplices (block-diagonal Fisher information). Thus $d \in \{20, 42, 72\}$ for $K \in \{5, 7, 9\}$.

2.2 Fisher–Rao embedding

Each row of $T(N)$ defines a point on the $(K - 1)$ -simplex. The Fisher information metric on the multinomial simplex in free coordinates $\theta = (p_1, \dots, p_{K-1})$ is $G_{ab} = \delta_{ab}/p_a + 1/p_K$. The full $K \times K$ transition matrix lives on a product of K such simplices, giving a d -dimensional Riemannian manifold with block-diagonal Fisher metric. Two manifold quantities are computed for each (readout, N) pair: $\log \det G(N)$ (log-determinant of the Fisher metric tensor at $T(N)$), and `fisher_speed_dn` (Fisher path speed $\|d\theta/dN\|_G$, estimated via finite differences between consecutive $T(N)$ matrices).

2.3 Exponential relaxation fit

For each readout, the Fisher speed profile $v(N)$ is fitted to two models: exponential $v = v_\infty + A \exp(-N/\tau)$, and power law $v = v_\infty + aN^{-b}$. Model selection is by AIC. The product $A \times \tau$ is recorded for each readout. The grammar threshold N^* is taken from the Fisher trace maximum on the same N -grid.

3 Results

3.1 Exponential dominance

The exponential model is preferred (lower AIC) in 25–26 of 28 readouts for all three K values: 26/28 at $K=5$, 25/28 at $K=7$, 25/28 at $K=9$. The Fisher path speed universally shows high values at small N (rapid exploration of the manifold) followed by monotonic decay toward a near-zero asymptote.

3.2 Relaxation timescale and threshold

At $K=7$, the median relaxation timescale is $\tau \approx 22,400$ (range 13,800–39,700). The ratio τ/N^* has median 3.22, meaning N^* sits at $N^*/\tau \approx 0.31$ on the exponential curve—where only $\sim 27\%$ of the total decay has occurred. N^* thus marks the onset of detectable structure, not the completion of relaxation.

[Figure 1 placeholder] Fisher path speed (blue) and $\log \det G$ (red) vs. N for 28 readouts ($K=7$). Dashed vertical lines: N^* .

Figure 1: Fisher path speed (left axis) and $\log \det G$ (right axis) versus N for all 28 Sycamore readout configurations ($K=7$). Dashed vertical lines indicate N^* .

[Figure 2 placeholder] Phase-transition style panels (trace, KL, entropy, Frobenius).

Figure 2: Fisher information phase transition analysis for 28 Sycamore readouts ($K=7$).

3.3 $A \times \tau$ stability within K

Table 1: $A \times \tau$ across topologies ($K=7$, Sycamore, exponential-fit readouts only).

Topology	n	Median $A \times \tau$	Std	CV
1D Snake	6	66.04	15.32	0.23
2D Block	13	65.87	9.70	0.15
Bulk Full	6	64.95	8.47	0.13
All	25	65.87	10.73	0.16

Table 1 summarizes $A \times \tau$ for $K=7$. The median is 65.87 across all 25 exponential-fit readouts. The three topologies agree within $\sim 2\%$ in median $A \times \tau$: 1D Snake 66.04, 2D Block 65.87, Bulk Full 64.95. This suggests that $A \times \tau$ is largely a property of the manifold geometry and the learning pipeline, not of the physical hardware topology.

For $K=5$ (26 exponential-fit readouts) topology-level medians of $A \times \tau$ are 54.13 (Snake, $n=7$), 49.00 (Block, $n=13$), and 56.79 (Bulk, $n=6$), with $\text{CV}=\text{std}/\text{median}$ of 0.22, 0.16, and 0.11. For $K=9$ (25 exponential-fit readouts) the corresponding medians are 75.31, 73.50, and 74.30 ($n=7, 13, 5$) with CV 0.16, 0.17, and 0.07. Thus within- K stability is not special to $K=7$; Snake rows are typically slightly noisier than Bulk in this CV sense.

Table 2: $A \times \tau$ scaling across K (28 Sycamore readouts; exponential fits).

K	$d=K(K-1)$	n_{exp}	Median $A \times \tau$	Pred.*	Error
5	20	26	53.32	53.64	0.6%
7	42	25	65.87	64.94	1.4%
9	72	25	74.30	74.90	0.8%

*Power law $A \times \tau = 21.5 K^{0.57}$ fit to the three medians; all residuals $< 1.4\%$.

[Figure 3 placeholder] Median $\pm$ std of $A \times \tau$ vs. K with power-law overlay.

Figure 3: $A \times \tau$ (median $\pm$ std across readouts) versus $K \in \{5, 7, 9\}$.

3.4 Power law scaling across K

3.5 LLM entropy series (exploratory)

The same analysis applied to LLM entropy time series (Qwen2.5-Coder:7B, $K=7$, twelve conditions) yields noisier profiles: the power-law tail model is preferred by AIC in *all twelve* conditions, consistent with multi-scale temporal structure documented in the Curvature Research Note. We therefore do not treat exponential sub-parameters (τ , A , hence $A \times \tau$) as primary estimands for LLMs; the reported median $A \times \tau \approx 91.5$ ($p=0.11$ vs. 65.97) is exploratory only and is excluded from the main Sycamore scaling analysis.

4 Comparison with the Coppo–Pranzini–Verrucchi threshold

4.1 The macroscopicity threshold

Coppo, Pranzini and Verrucchi (2022) derived a threshold system size $N_t(\varepsilon, \delta)$ beyond which a quantum system admits a classical-like description via generalized coherent state (GCS) discrimination. For the $\mathfrak{su}(2)$ case: $N_t = \ln(\varepsilon)/\ln[\cos(\delta/2)]$, where ε is a tolerance and δ is experimental resolution. The GCS overlap decays exponentially in N .

4.2 Functional form comparison

The overlap speed is a pure exponential in the Coppo construction; Grammar Fingerprinting uses $v(N) = v_\infty + A \exp(-N/\tau)$. Both are exponential relaxations with a single characteristic timescale. Threshold dependence on tolerance is logarithmic in both constructions.

4.3 Two thresholds on one curve

The Grammar Fingerprinting threshold N^* sits at $N^*/\tau \approx 0.31$ (onset of structure). The Coppo threshold N_t (at $\varepsilon = 0.22$, their example) sits at $N_t/\tau \approx 3.03$ (classical completion). Their ratio is $N_t/N^* \approx 9.7$ for $\varepsilon = 0.22$.

[Figure 4 placeholder] Normalized decay with N^* and N_t markers.

Figure 4: Normalized exponential decay $v/v_0 = \exp(-N/\tau)$ with N^* and N_t marked.

4.4 What matches, what does not

The functional form (exponential relaxation $\rightarrow$ logarithmic threshold dependence) matches at a qualitative level. The within- K stability of $A \times \tau$ across hardware topologies mirrors the algebra-independence of the Coppo ε -dependence (always proportional to $\ln \varepsilon$). However, the Coppo invariant $A_C \times \tau_C = 1/2$ on S^2 does not hold for Grammar Fingerprinting ($A \times \tau \approx 66$ at $K=7$, scaling $\propto K^{0.57}$ across the three K values). This is expected: Coppo et al. work on the $SU(2)$ coherent-state manifold S^2 , while Grammar Fingerprinting operates on a product of $(K-1)$ -simplices. The correspondence is structural (shared functional form and threshold logic), not parametric.

5 Discussion

The main finding is universal relaxation of the learned transition matrix toward a fixed point on the Fisher–Rao manifold, exponentially in N , with τ systematically $\sim 3\times$ larger than the Fisher trace threshold N^* . The product $A \times \tau$ is an empirical pipeline summary at fixed K , largely insensitive to hardware topology (Table 1 and the $K=5, 9$ paragraph above), and scales sublinearly in K over $\{5, 7, 9\}$.

The Coppo comparison supplies a theoretical frame: exponential distinguishability decay and logarithmic tolerance thresholds. Whether the analogy reflects a deeper formal connection remains open.

The sublinear K -scaling ($A \times \tau \sim K^{0.57}$) requires further investigation: three K values identify sublinearity but not a unique functional form. Additional K values would test robustness of the exponent.

6 Summary

1. Fisher path speed decays exponentially in N , with the exponential model preferred in 25–26/28 readouts for each $K \in \{5, 7, 9\}$.
2. τ is systematically $\sim 3.2\times$ larger than N^* , placing N^* at the onset ($\sim 27\%$ decay) of exponential relaxation.
3. $A \times \tau$ is stable across hardware topologies within each fixed K among exponential-fit readouts (Table 1 at $K=7$; qualitatively similar at $K \in \{5, 9\}$). Across $\{5, 7, 9\}$, median $A \times \tau$ scales as $21.5 K^{0.57}$ (residuals $< 1.4\%$; Table 2).
4. The exponential relaxation form parallels the Coppo–Pranzini–Verrucchi macroscopicity construction; N^* and N_t mark different milestones on a decay curve.
5. A formal bridge between Fisher–Rao geometry of transition matrices and the Page–Wootters mechanism remains an open problem.

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Data and code

Sycamore readout data: Dryad DOI 10.5061/dryad.k6t1rj8. Code and results: <https://github.com/csaplard/Quantum>

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